Worksheet 3B: Quadratics — Factoring and Solving

Part A: Factoring — The Reverse of FOIL (Warm-up)

In Worksheet 3A, you expanded $(t+3)(t+4)$ to get $t^2+7t+12$.

Now we go backwards: given $t^2+7t+12$, find the two binomials.

The Key Insight:

To factor x^2+bx+c, find two numbers that: - **Multiply** to give c (the constant term) - **Add** to give b (the coefficient of x)

Example: Factor $x^2+7x+12$

- Need two numbers that multiply to 12 and add to 7
- Factors of 12: (1,12), (2,6), (3,4)
- Which pair adds to 7? $\rightarrow$ 3 and 4
- Answer: $(x+3)(x+4)$

Factor each expression. Check your answer by FOILing!

1. t^2+5t+6

- Multiply to _____, add to _____
- Numbers: _____ and _____
- Factored: (_____)(_____)

2. $p^2+8p+15$

3. $m^2+9m+20$

4. $r^2+11r+24$

5. $k^2+7k+10$

6. $n^2+12n+35$

Part B: Factoring with Negative Middle Terms (8 problems)

When the middle term is negative but the constant is positive, both numbers are negative.

Example: Factor $x^2-7x+12$

- Need two numbers that multiply to +12 and add to -7
- Both must be negative (negative $\times$ negative = positive)
- Numbers: -3 and -4
- Answer: $(x-3)(x-4)$

Factor completely:

7. $t^2 - 5t + 6$

8. $p^2 - 9p + 14$

9. $m^2 - 8m + 15$

10. $r^2 - 10r + 21$

11. $k^2 - 11k + 28$

12. $n^2 - 13n + 36$

13. $w^2 - 6w + 9$ — *What's special about this one?*

14. $s^2 - 14s + 49$ — *Another special one!*

Part C: Factoring with Negative Constants (10 problems)

When the constant term is negative, one number is positive and one is negative.

Example: Factor $x^2 + 2x - 15$

- Need two numbers that multiply to -15 and add to +2
- One positive, one negative
- Factors of 15: (1,15), (3,5)
- Which pair gives +2 when combined? $\rightarrow$ +5 and -3
- Answer: $(x + 5)(x - 3)$

Factor completely:

15. $t^2 + 3t - 10$

16. $p^2 + 2p - 8$

17. $m^2 - 2m - 15$

18. $r^2 - 3r - 18$

19. $k^2 + 4k - 21$

20. $n^2 - 5n - 24$

21. $w^2 + w - 30$

22. $s^2 - 6s - 27$

23. $h^2 + 7h - 44$

24. $u^2 - u - 56$

Part D: Special Factoring Patterns (10 problems)

Pattern 1: Difference of Squares

$$a^2 - b^2 = (a + b)(a - b)$$

Look for two perfect squares separated by subtraction.

Factor:

25. $t^2 - 25$

26. $p^2 - 81$

27. $m^2 - 100$

28. $4r^2 - 9$

29. $16k^2 - 49$

30. $25n^2 - 1$

Pattern 2: Perfect Square Trinomials

$$a^2 + 2ab + b^2 = (a + b)^2$$
$$a^2 - 2ab + b^2 = (a - b)^2$$

Check: Is the middle term exactly twice the product of the square roots of the first and last terms?

Factor:

31. $w^2 + 10w + 25$

32. $s^2 - 12s + 36$

33. $h^2 + 16h + 64$

34. $4u^2 + 12u + 9$

Part E: Factoring When the Leading Coefficient 1 (12 problems)

When you have $ax^2 + bx + c$ with $a \neq 1$, factoring is trickier.

Method: AC Method (Factor by Grouping)

Example: Factor $2x^2 + 7x + 3$

1. Multiply $a \cdot c$: $2 \cdot 3 = 6$
2. Find two numbers that multiply to 6 and add to 7: $\rightarrow$ 6 and 1
3. Rewrite the middle term: $2x^2 + 6x + 1x + 3$
4. Group and factor: $2x(x + 3) + 1(x + 3)$
5. Factor out common binomial: $(2x + 1)(x + 3)$

Factor completely using the AC method:

35. $2t^2 + 5t + 3$

- $a \cdot c =$ ____
- Numbers that multiply to ____ and add to 5: ____ and ____
- Rewrite: $2t^2 +$ ____$t +$ ____$t + 3$
- Factor: (____)(____)

36. $3p^2 + 10p + 8$

37. $2m^2 + 9m + 4$

38. $5r^2 + 13r + 6$

39. $3k^2 - 7k + 2$

40. $2n^2 - 11n + 5$

41. $4w^2 + 4w - 3$

42. $6s^2 - 7s - 3$

43. $3h^2 - 10h - 8$

44. $4u^2 - 25$ — *Difference of squares!*

45. $9v^2 + 12v + 4$ — *Perfect square trinomial!*

46. $2q^2 - 18$ — *Factor out GCF first!*

Part F: Always Look for GCF First! (6 problems)

Before using any other technique, always check for a **Greatest Common Factor**.

Example: Factor $3x^2 + 12x + 9$

1. GCF of 3, 12, 9 is 3
2. Factor out 3: $3(x^2 + 4x + 3)$
3. Factor the trinomial: $3(x + 1)(x + 3)$

Factor completely (GCF first, then continue):

47. $2t^2 + 10t + 12$

48. $5p^2 - 20p + 15$

49. $3m^2 - 27$

50. $4r^3 + 8r^2 - 12r$

51. $6k^2 - 24k + 24$

52. $2n^3 - 8n^2 - 10n$

Part G: Solving by Factoring — Zero Product Property (12 problems)

The Zero Product Property:

If $A \cdot B = 0$, then $A = 0$ or $B = 0$ (or both).

This lets us solve factored equations!

Example: Solve $x^2 - 5x + 6 = 0$

1. Factor: $(x - 2)(x - 3) = 0$
2. Apply Zero Product Property: $x - 2 = 0$ or $x - 3 = 0$

3. Solve each: $x = 2$ or $x = 3$

Solve by factoring:

53. $t^2 + 7t + 12 = 0$

54. $p^2 - 6p + 8 = 0$

55. $m^2 + 2m - 15 = 0$

56. $r^2 - 9 = 0$

57. $k^2 - 10k + 25 = 0$ — *How many solutions?*

58. $2n^2 + 5n - 3 = 0$

59. $3w^2 - 7w + 2 = 0$

60. $s^2 + 8s = 0$ — *Don't forget: s is a common factor!*

61. $h^2 = 16$ — *Rearrange first!*

62. $u^2 - 3u = 10$ — *Rearrange to standard form first!*

63. $2v^2 + 6v = -4$ — *Rearrange, then factor out GCF*

64. $t(t - 4) = 12$ — *Expand first, then rearrange*

Part H: Completing the Square (10 problems)

Sometimes factoring doesn't work nicely. **Completing the square** always works.

The Idea: Turn $x^2 + bx$ into a perfect square by adding $\left(\frac{b}{2}\right)^2$

Example: Solve $x^2 + 6x = 7$

1. Take half of 6, square it: $\left(\frac{6}{2}\right)^2 = 9$
2. Add 9 to both sides: $x^2 + 6x + 9 = 7 + 9$
3. Factor left side: $(x + 3)^2 = 16$
4. Take square root: $x + 3 = \pm 4$
5. Solve: $x = -3 + 4 = 1$ or $x = -3 - 4 = -7$

Solve by completing the square:

65. $t^2 + 4t = 5$

- Half of 4, squared: _____
- Add to both sides: $t^2 + 4t +$ _____ $= 5 +$ _____
- Factor: $(t +$ _____$)^2 =$ _____
- Solutions: $t =$ _____

66. $p^2 - 8p = 9$

67. $m^2 + 10m = -21$

68. $r^2 - 2r = 15$

69. $k^2 + 6k + 5 = 0$ — *Move 5 to the right first*

70. $n^2 - 4n - 12 = 0$

71. $w^2 + 3w = 4$ — *Yes, you can complete the square with odd numbers!*

72. $s^2 - 5s = 6$

73. $h^2 + 8h + 20 = 0$ — *What happens here?*

74. $u^2 - 6u + 10 = 0$ — *And here?*

Part I: The Quadratic Formula (10 problems)

For any quadratic $ax^2 + bx + c = 0$:

$$\boxed{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$$

Example: Solve $2x^2 + 5x - 3 = 0$

- $a = 2$, $b = 5$, $c = -3$
- $x = \frac{-5 \pm \sqrt{25-4(2)(-3)}}{2(2)} = \frac{-5 \pm \sqrt{25+24}}{4} = \frac{-5 \pm \sqrt{49}}{4} = \frac{-5 \pm 7}{4}$
- $x = \frac{-5+7}{4} = \frac{1}{2}$ or $x = \frac{-5-7}{4} = -3$

Solve using the quadratic formula:

75. $t^2 - 5t + 6 = 0$ — *Verify by factoring too!*

76. $p^2 + 3p - 10 = 0$

77. $2m^2 + 7m + 3 = 0$

78. $3r^2 - 2r - 1 = 0$

79. $k^2 - 4k + 1 = 0$ — *Answer won't be a nice integer*

80. $n^2 + 2n - 2 = 0$

81. $w^2 + 4w + 5 = 0$ — *What happens under the square root?*

82. $2s^2 - 3s + 5 = 0$ — *Same question!*

83. $h^2 - 6h + 9 = 0$ — *What's special about this one?*

84. $4u^2 + 4u + 1 = 0$

Part J: The Discriminant — Predicting Solutions (8 problems)

The expression under the square root, $b^2 - 4ac$, is called the **discriminant**.

It tells you what kind of solutions to expect:

Discriminant	Number of Real Solutions
$b^2 - 4ac > 0$	Two different real solutions
$b^2 - 4ac = 0$	One real solution (repeated)

Discriminant	Number of Real Solutions
$b^2 - 4ac < 0$	No real solutions (complex numbers)

Without solving, calculate the discriminant and predict the number of real solutions:

85. $t^2 + 5t + 6 = 0$

- $b^2 - 4ac = 25 - 24 = 1$
- Discriminant is _____ (positive/zero/negative)
- Number of real solutions: ____

86. $p^2 - 8p + 16 = 0$

87. $m^2 + 2m + 5 = 0$

88. $r^2 - 3r - 10 = 0$

89. $k^2 + k + 1 = 0$

90. $3n^2 - 2n - 1 = 0$

91. $4w^2 - 12w + 9 = 0$

92. $2s^2 + s + 1 = 0$